Meth. Lorraine

A Sketchy critique of Part of
St. Beghris Matins.

S B.

Spelling careless. Symptom with the Printer

Thesis for Edinburgh University

by

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Goitre

Goitre has of late years become to be much better understood than formerly, both as regards its causation and its treatment. It is not difficult to understand that regions or countries may present differences in the diseases to which they are generally liable, or that morbid conditions fundamentally common to all, should manifest characteristic peculiarities in each; but that a disease should manifest itself in given spots of a district without any satisfactory, perceptible & external causes to account why it should select such a spot in preference to another was to a certain extent a mystery —

The diversities of climate, varying from the extreme cold of the Northern regions, to the intense heat of Tropical Countries; the countless & easily recognisable differences of the atmosphere involving various degrees of humidity, dryness, rarity &c, which these different territories present the inhabitants of these regions also differing so much in their moral and physical conditions, account satisfactorily & fully for the former but for endemic diseases generally & goitre in particular no such evident reason exists.

It infests all climates, hot & cold, every variety of atmosphere, moist and dry, & peoples of all colours and tongues — And coming down from such

such vast generalities to less extensive ones we still find goitre not more invidious in these than in those. – It is developed in no particular spot of country more than another, so far as the relative positions of these spots to the sea are concerned, infecting the fens of Norfolk and the comparatively higher lands of Yorkshire as well as the mountainous Alps. & has been found on one side of a ravine in these mountains while on the other side it did not exist; nor does the relative habits or industry of the inhabitants of localities seem to cause it, to maintain it or to promote its severity, for one locality is infested, while another similar in these and other respects is not, attacking in widely different localities the indolent Hindoo & the active agricultural peasant in Derbyshire, attacking affluence & ease as well as honest & laborious employment – In fact look as we like at the external circumstances of an infected locality, for anything like a satisfactory reason for its prevalence, either as regards that locality itself or the people which inhabit it, we look in vain, because we can turn to another nearly, it may be precisely, equal to it, which is as evidently free from goitre as the other is not. Its being endemic to certain localities has given rise to successful efforts to discover the cause of its prevalence in these districts.

Some, poisonous influence, similar to malaria of the

the atmosphere was first suspected as the cause & no doubt the insalubrity of the air in some places where goitre is abundant, has its influence for evil in this particular, as over other diseases, but further than in its probable maintainance & severity no more can be said of the air being the cause of goitre; in proof of this, it exists endemically in many of the most salubrious districts in this country, districts to which annually in the summer many repair & even in the winter in many cases from the salubrity of the climate alone. And even taking an opposite view of the matter, we equally find in the most notoriously unhealthy atmospheres no goitre endemically prevailing. Of course unhealthy atmosphere generate disease, as ague & other fevers are caused by breathing the atmosphere of malarious districts, each malaria having its own particular type of disease, but so far as we are aware no goitre has been found to exist where such deleterious atmosphere exists, without at least the endemic development of disease peculiar to such malaria — In other words, there is no malarious atmosphere, so far as is known which developes goitre as a desease, sui generis.

The researches of McClelland in India completely disprove of the opinion that the atmosphere has anything to do with the

the causing of goitre in infected places — He finding this disease in ~~one~~ one part of a district while in another of the same district it was absent, while so far as he could discern, there appeared no inequality in the physical circumstances affecting each — In this country goitre exists in places where there is the greatest diversity of the nature of the atmosphere.

Popular belief in infected localities condemns the water which they use as the cause of this malady and the researches of many tend completely to corroborate this belief — The inhabitants deem themselves to be living in a very unhealthy situation and express as their sole reason for this that the water is sandy; such is frequently their expression for what is more commonly termed hard. That this is the case is amply corroborated by M^c^Clelland who found as before said, goitre in one part of a district while in another of the same district and subjected to the same physical conditions, goitre did not exist and he attributes this to the presence in the water of the salts of lime — He found even amongst the inhabitants of the same village containing three castes a most marked corroboration, two castes, respectively named Rajpoots and Domes who used hard water were almost wholly suffering from goitre while the other named Brahmins who used different water were as entirely free

what cause can we ascribe the immunity of one caste of the inhabitants of this village, and the almost universal affection of the other two— Difference of caste does not here imply a difference of pecuniary circumstances and consequently of the comforts of life; In these respects the three castes in this village are on perfect equality— The village is raised about one hundred feet above the level of the valley, and the mountain at the foot of which it is situated rises with a gentle slope and is not in this vicinity at all rugged. It is chiefly composed of transition limestone and the village is erected on a conglomerated rock composed of calcareous tuff inclosing fragments of Rock. There is a spring in the valley about one hundred yards from the village, bearing on its first appearance the character of a mineral spring. The water bursts forth with strong ebullition and agglutinates the sand and gravel by which it is surrounded, by the deposition of calcareous Tuff— The temperature & quantity of the water is the same at all seasons— The former inhabitants of the village, aware perhaps of the noscious effects of this spring, had an aqueduct formed, by which water is conveyed into the Brahmin portion of the village from a distant source— The aqueduct having been suffered to get out of repair, the quantity of water it

it transmits is reserved exclusively for the Brahmins except during the rainy season when the water being plentiful, the Rayroots also use that of the aqueduct, but the Domes have no alternative at any season but to use the water from the Springs. McClelland asserts that during his investigations over a very extensive country, he did not in one instance find a locality infected to any extent with goitre which was not supplied with water, which percolated through limestone rocks. Numerous other testimonies, with equally decisive views, from equally successful and talented investigators could be adduced in favour of the water used being the proximate cause of Goitre, but it is needless to keep repeating in the words of other authors, what has been so emphatically expressed by M'Clelland — It appears to me that too little attention in this, as in other matters, has been paid to the Vox Populi of goitrous districts. Although probably they do not point expressly to goitre as the immediate result of the water daily used by them, yet they do unequivocally condemn the water as the cause of those ailments which present a train of symtoms which, we shall subsequently see, frequently unless checked result in the development of Goitre — And it is with a view of obviating these effect that the inhabitants in many instances, boil all the water they intend using, which we know renders the water soft

Soft by precipitating the calcareous salts—
It is a matter of considerable moment that this prophylactic method should be more generally known & acted upon ——

Dr Begbie in his article on Anaemia says. "That an anaemic or spanaemic condition of the system is the true cause of Bronchocele in those districts of the country where it prevails in a endemic form, I am not prepared to assert; but the wretched condition of a vast number of its unhappy victims, and the physical causes which have generally been assigned for its prevalence, together with the pale sanguineous ~~countenance~~ complexion, the protuberant eyeballs, and other signs of anaemia manifested by its subjects, render the conjecture not improbable — It is a curious coincidence and not unworthy of remark, that the blood impoverished coalminers of Anzain and the goitrous Peasant of the Alps, both attribute their apparently dissimilar maladies to the quality of the water they are accustomed to use for their daily sustenance, and that the late researches of McClelland, Bally and Inglis, in different and widely separated parts of the world, have well nigh established the fact, that the endemic prevalence of goitre is connected with the use of water impregnated with calcareous salts. May not the habitual employment of such water, so influence digestion, assimilation & sanguification as to lead to the anaemic condition, which in its turn again

again, is the cause of the developement of goitre" — We believe that the suggestion conveyed in the latter part of the quotation from Dr Begbie will ultimately turn out the correct one — It would be an interesting subject of investigation to the Physiological Chemist, how the long continued use of water impregnated with salts of lime might effect the process of nutrition, not merely in the restricted meaning of nutrition, but in its more extended sense, involving assimilation, secretion & excretion — Although for obvious reasons interference with desestion or chylification must of necessity involve the whole process, yet it is possible that hard water or the absence of free Carbonic Acid which it presupposes, may have a more remote effect on the functions of the blood itself —

Being at present located in a part of country where goitre prevails, I have incidently seen a few cases of goitre and in all without exception their appearance and history, so far as I could learn suggested to me the certainty of their labouring under some form of malnutrition or other equally efficacious cause of an impoverished state of the blood — It did appear to me also that whatever caused the goitre, caused also the anaemic condition, but my ideas on the subject were somewhat vague, until about a month ago, I happened in with Dr Begbie's Book entitled "Contributions to Practical Medecine and

on reading the subject Anaemia & its Consequences I was much impressed with its statements. Whatever may be the result of future enquiries into this subject, which no doubt will be induced by this able Paper and coming from the pen of one whose deliberate expression of Opinion, founded on great experience and deep research, must always be attended with mighty interest to the profession I think it may be conceded that Dr Begbie has presented conclusions drawn from established premises, which account the most satisfactory of anything hitherto, for the manifestations of which his paper treats; and conclusions which will take no little experience and ability to confute.

A case has been partially reported to me by an intelligent practitioner in this vicinity, in which his advice was sought for an enormous goitre. I have seen the woman & from her own statement as to its origin, it would appear to be one which thoroughly tends to substantiate the hint set forth by Dr Begbie — The case is that of a woman about 40 years of age, who has borne a pretty large family, and possessed previously to this attack moderately good health. It commenced with nausea and vomiting with frequent severe Colic, constipation alternating with diarhoea attended also with loss of apetite, upon these supervened uterine disorder, and from the

Menorrhagia

She was led to suppose herself at her change of life. After these symtoms she first noticed the swelling in her neck & which was developed so rapidly as to necessitate her to obtain the services of the practitioner alluded to. She was put upon a tonic treatment and the size of the goitre was lessened (so she says) by one application of the ointment she had received & which was the Biniodide of Mercury ointment recommended by Capt Cunningham — She recovered from the symtoms described, but the tumour, much lessened however, still remains and tho reverse of being at her change of life, she is at the present moment pregnant.

In this case we first find symtoms, caused, it may be, by the water, not necessarily however which develope morbid uterine symtoms, causing a constant and exhausting drain upon the blood, already impoverished as it must have been by the Malassimilation in the first instance and subsequently inducing a state causing the absence of matters to assimilate — It cannot be positively asserted that the nature of the water, which in this district possesses the characteristic hardness as in other goitrous situations, was the exciting cause, any more than any other known to produce the symtoms mentioned, but its invariable presence when goitre is endemic coupled with popular testimony in this respect

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gives an amount of evidence equal almost to positive proof — Absolute proof must be found by the Physiologist and chemist, and it is possible when the attention of these are drawn to the subject, it may be found as suggested by Dr Begbie, that the long continued use of hard water, has an unperceivable action on the healthy acidity of the gastric juice, imperceptible at first, but gathering strength as other conditions are brought into play, until at last indigestion & malassimilation are the result, as in the case stated — The continued enlargement of the Thyroid gland seems to be a degeneration of the vascular variety to the more persistent kind, probably ensuing from the exudation & organisation of lymph, while in the state of excessive congestion which first characterised it.

A short time ago, the Medical Gentlemen before mentioned, was examining a young woman in my presence, whose ulcerated tonsils & apthous condition of the mucous membrane of the mouth, combined with other symptoms; proclaimed a disordered Stomach & digestion, said that this case was such a one, in which he would expect Goitre to be developed if it had been allowed to go on unchecked — This opinion did not seem to me at the time to be of much value, but there can be no doubt that it was his conviction that Bronchocele was somehow or other connected with disordered digestion and assimilation

and I think it of some consequence as proceeding from one whose practice has been extensive in the district for a long series of years.

There is another case, very similar to Dr Begbie's & illustrative of his views, which has come under my own observation, it is that of a woman past the prime of life whose husband I have been attending for a considerable period, and who invariably upon going to see her husband treated me to a recital of her own peculiar ailments. I had her previous history from some of the neighbours, and was warned against her complainings and talkativeness - I consequently never attended to her, nevertheless I could not help discerning her staring eyes, her thin and pale appearance and peculiar agitated, nervous manner. She told me over and over again of what she called her thin blood, ringing ears, her palpitation and indigestion & her incapability of taking any food, even tea without her customary draught of Chamomile infusion and her utter abhorrence of anything like animal food —

One day dropping in upon her rather suddenly, I spoke without her seeing me. She turned to me with such a frightened look and nervous agitated manner, that I was astonished, more so at the manner in which she stared, her eyes literally appeared bursting from the Sockets — I observed now for the first time a goitre evidently enlarged by the agitation she was in, as it lessened in size

size

as she calmed down. Considering the number of times I had been inflicted with a recital of her ailments. I was surprised she had not before mentioned this swelling in her neck, as she had repeatedly told before of fits of dimness of vision & on asking her why she had never mentioned it, she replied that it was only a short time since she had noticed it and that its decreasing at times very much made her hope it would ultimately go away. That it created so little inconvenience, as to have forgotten about it when she got nervous however to which she said she was very subject. it got much larger.
I heard now for the first time of a profuse leucorrhoea to which she had been subject for a lengthened period. This case presents features very similar to Dr Begbie's, and of this cause I entertain no doubt. The existence for a long time of disordered nutrition, with a uterine disorder, whether succeeding or preceding the former I could not clearly elicit, peculiarly adapted for producing what she very aptly termed thin blood, and upon this anaemic condition ~~again~~ gradually increased by the prolonged duration of the causes, supervene as symptoms the exophthalmos and goitre.

The views so intelligibly expressed by Dr Begbie in his work relating to this subject are regarded as fallacious by Professor Laycock, in an ingenious paper recorded in the Edinburgh Medical Journal of February last, although it professes to combat

on

one
items specially of the results arrived at by Dr Beyle
Still it appears to me, that, that one is so
intimately interwoven with the results as a whole,
that it is impossible to compute it without materially
damaging Dr Beyle's whole conclusions.

Dr Laycock begins with remarks upon the instability
of the more common symtoms of anaemia and
thence he proceeds to show that those more rare
ones, viz exophthalmos & Goitre, are so, from their
not depending upon anaemia at all, but upon
some defect or lesion of the nervous system
I apprehend however that the mere absence of one or
more of the more common symtoms of anaemia is
what may be expected from the nature of the disease,
it is not to be regarded as a criterion, from which to
conclude, its presence or absence as a disease.

Pure anaemia is a diminution of the red corpuscles
of the blood, from their healthy ratio to the other
constituents; arising either from causes operating
immediately on the blood, as haemmorhage, or what
is equivalent, profuse and tedious discharges,
and thereby creating an abnormal consumption
of the red corpuscles, which cannot be normally
supplied, or on the contrary arising from causes
whose operation immediately impedes the supply
or further these may be united, an abnormally
small supply with an abnormally increased
demand — Thus anaemia is a deterioration
of the quality of the blood — an improper
development

developement or degeneration into some lower organization which does not functionally like them.

As all parts of the body are alike dependent at least for the material, if not for the influence, wherby their several functions are performed, we might naturally expect a gradual and equal nondevelopement of function or vitality, but, so far as I am aware, it is not so in almost all instances of this disease, but according to individual peculiarities or cachexias, symtoms attached to systems, tissues, or organs of the body take the lead to the exclusion or rather subordination of the others.

According to M. M. Andral and Gavarret, well marked cases of Anaemia present a diminution of at least 3/4 of the red corpuscles.

I am apprehensive it would be a difficult matter to maintain, that the symtoms which marked a diminution of 3/4 of the corpuscles, and allowing for intensity, are still the same in character or even, as it were, built upon the symtoms which marked a diminution of 4/4 — As the anaemic murmur of the heart and large vessels is considered a symtom of an advanced stage of the disease & as Goitre & exopthalmos when they do exist, are in most instances associated with it, it may be fairly taken for granted, that they also are advanced symtoms of the anaemic condition.

It may be even reasonably surmised that goitre and exopthalmos are developed when the corpuscles

have been diminished to that extent as to occasion the sounds referred to and are dependent to some extent, or even wholly, upon the conditions giving rise to the latter symptoms.

The comparative rarity of goitre & exophthalmos as symptoms of the anaemic condition is accounted for on these grounds, that they are advanced stages of the disease & that the urgency of their case, cause parties to obtain relief before they are developed.

I am afraid Dr Laycock does not make out his case so clearly as Dr Begbie, and it would be something more intelligible, something more tangible if it were admitted by Professor Laycock, that a depressed nervous vitality consequent on some cachexia influenced the anaemic condition, and through this the goitre and exophthalmos, or that the anaemic condition so produced, reacted on the nervous system, causing the desired nervous lesion to account for the exophthalmos and goitre —

In fact Dr Laycocks paper involves, what is very indistinctly understood, the natural & mutual relation of the nervous & vascular systems, and if it is abrogated for the nervous system functions, involving the highest vital processes, a sort of comptroller of the household, in that case A may be right.

Not- the least effective part of Dr Begbies remarks is the manner in which he accounts for the predominance of this disease in women over men

It has been attempted to account for this ways without number, but never once in so simple and effective a manner as that presented by Dr Begbie The treatment too, which proposes treating Goitre in the same way as for Anaemia promises to be as successful as it is rational.

I think we may conclude that, Endemic Goitre is developed from Anaemia caused by the quality of the water used, acting in an unhealthy manner on nutrition.

The way goitre may supervene on Anaemia may be considered as undecided: but it is possible that the attenuated blood, giving rise to abnormal sounds & diseased action of the heart, acts in such a manner on the branches of the Sympathetic, as to be reflected to the branches of the same nerve to the Thyroid Gland & that this is rendered more probable by the fact that the size of the gland is increased when the excitement of the heart is increased,

That persistent Bronchocele is increased bulk of the gland caused by previous congestion which may be alleviated after removal of the Anaemic condition by those means which promote absorption especially the ~~Hydriodate~~ Biniodide of Mercury ointment.

W. Lorraine